Coffee antioxidants protect against liver damage in Non-alcoholic Fatty Liver Disease (NAFLD)

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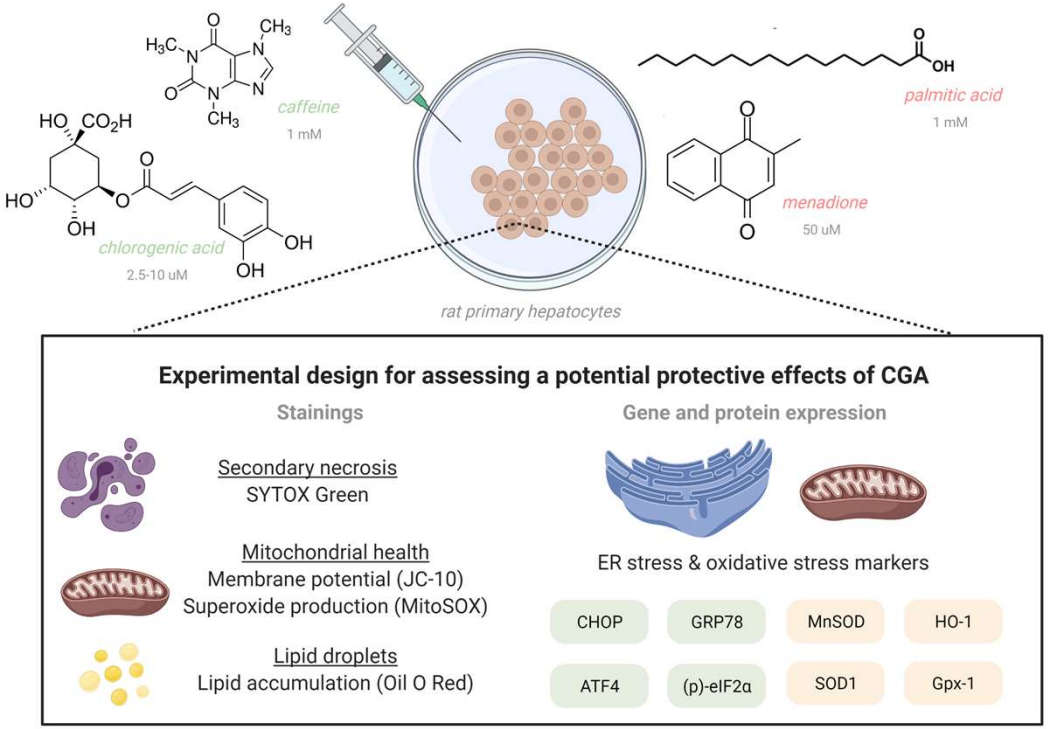
Introduction & aim

Non-alcoholic fatty liver disease (NAFLD) has become the leading liver disease in the world with an estimated prevalence of 25% in the global population[1,2,3]. Despite this, there is currently no pharmacological therapy to prevent NAFLD development and progression. Regarding the pathophysiology of NAFLD, it is known that hepatic lipid accumulation, specifically saturated fatty acids (SFAs), triggers oxidative stress (OxS) and depletes antioxidant defenses in the liver. This causes lipotoxicity via mechanisms such as ER stress, mitochondrial dysfunction and the activation of cell death pathways[5]. Reactive oxygen species (ROS) play a key role in this as prolonged ER stress causes a substantial increase in intracellular ROS production and the disturbance of the mitochondrial membrane potential further aggravates the redox imbalance[6, 7]. Therefore, antioxidant compounds could be useful against lipotoxicity and as a potential therapy for NAFLD.

Epidemiological studies indicate that coffee consumption protects against chronic liver diseases, including NAFLD[8, 9]. Part of this protective effect can be explained by the high antioxidant potential of coffee, due to the presence of compounds such as caffeine (CF) and chlorogenic acids (CGA)[9]. Moreover, CGA has demonstrated some protection against lipotoxicity-induced ER stress and apoptosis[10].

The **present** study aims to examine caffeine and CGA's protective effects and explore whether antioxidants mediate CGA's hepatoprotective effect.

Material and methods[11]



Results

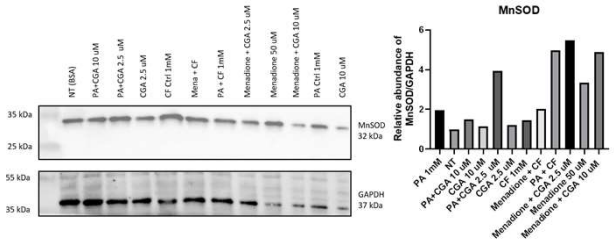


Figure 1: Caffeine (1 mM) and chlorogenic acid (CGA; 2.5 µM) increase antioxidant marker MnSOD in primary rat hepatocytes treated with palmitic acid.

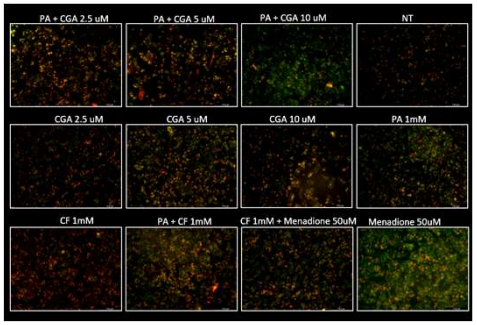


Figure 2 Caffeine and low concentrations of chlorogenic acid (CGA) prevent palmitate-induced depolarization of the mitochondrial membrane potential.

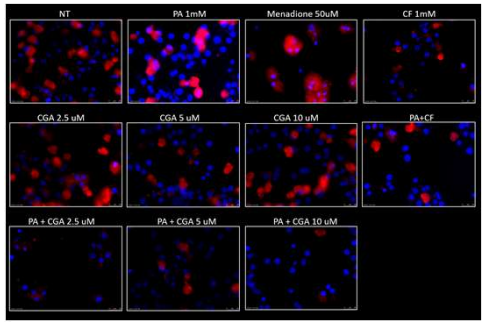


Figure 3: Chlorogenic acid prevents mitochondrial superoxide formation.

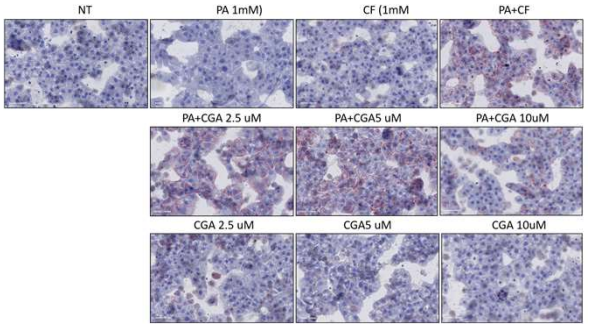


Figure 4 Coffee compounds caffeine and chlorogenic acid increase hepatic lipid accumulation.

Discussion & conclusion

Coffee compounds increase the mitochondrial antioxidant marker SOD2 (MnSOD) at protein level, but do not affect antioxidant nor ER stress markers at the mRNA level. In the mitochondria, a reverse dose-dependent effect of CGA on preserving the mitochondrial membrane potential (MMP) was observed with 2.5 µM CGA offering most protection and 10 µM displaying no protection. Also, caffeine reversed the palmitate-induced collapse of the MMP. In line with this, CGA reduced palmitate-induced generation of mitochondrial ROS production. In contrast, caffeine did not reduce palmitate-induced ROS generation. Interestingly, CF, CGA 2.5 µM or CGA 5 µM treatments are associated with higher lipid accumulation, suggesting a correlation between lipid droplet formation and the protective effects of coffee compounds against palmitic acid.

Future perspective

Antioxidant compounds in coffee may be vital as natural remedy against lipotoxicity described in NAFLD and related pathologies and can be considered both in the prevention as therapy of NALFD (and related diseases).

References

1. Friedman SL, Neuschwander-Tetri BA, Rinella M, Sanyal AJ. Mechanisms of NAFLD development and therapeutic strategies. Nature Medicine. Nature Publishing Group; 2018. pp. 908–922. doi:10.1038/s41591-018-0104-9
2. Angulo P. Medical progress: Nonalcoholic fatty liver disease. New England Journal of Medicine. N Engl J Med; 2002. pp. 1221–1231. doi:10.1056/NEJMra011775
3. Younossi ZM, Koenig AB, Abdelatif D, Fazel Y, Henry L, Wymer M. Global epidemiology of nonalcoholic fatty liver disease—Meta-analytic assessment of prevalence, incidence, and outcomes. Hepatology. 2016;64: 73–84. doi:10.1002/hep.28431
4. Younossi ZM. Non-alcoholic fatty liver disease – A global public health perspective. Journal of Hepatology. Elsevier B.V.; 2019. pp. 531–544. doi:10.1016/j.jhep.2018.10.033
5. Geng Y, Hernandez Villanueva AA, Faber KN, de Meijer VE, Blokzijl H, Moshage AJ. Lipotoxicity in non-alcoholic fatty liver diseases: mechanisms and clinical implications (unpublished). BBA Mol Cell Biol Lipids. 2019.
6. Malhotra JD, Kaufman RJ. The endoplasmic reticulum and the unfolded protein response. Seminars in Cell and Developmental Biology. Elsevier Ltd; 2007. pp. 716–731. doi:10.1016/j.semcdb.2007.09.003
7. Deniaud A, Sharaf El Dein O, Maillier E, Poncet D, Kroemer G, Lemaire C, et al. Endoplasmic reticulum stress induces calcium-dependent permeability transition, mitochondrial outer membrane permeabilization and apoptosis. Oncogene. 2008;27: 285–299. doi:10.1038/sj.onc.1210638
8. Setiawan VW, Wilkens LR, Lu SC, Hernandez BY, Le Marchand L, Henderson BE. Association of coffee intake with reduced incidence of liver cancer and death from chronic liver disease in the US multiethnic cohort. Gastroenterology. 2015;148: 118–125. doi:10.1053/j.gastro.2014.10.005
9. Marventano, S., Salomone, F., Godos, J., Pluchinotta, F., Del Rio, D., Mistretta, A., & Grosso, G. (2016). Coffee and tea consumption in relation with non-alcoholic fatty liver and metabolic syndrome: A systematic review and meta-analysis of observational studies. Clinical Nutrition, 35(6), 1269–1281.
10. Zhang Y, Miao L, Zhang H, Wu G, Zhang Z, Lv J. Chlorogenic acid against palmitic acid in endoplasmic reticulum stress-mediated apoptosis resulting in protective effect of primary rat hepatocytes. Lipids Health Dis. 2018;17: 270. doi:10.1186/s12944-018-0916-0
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